

```
#include <iostream> using namespace std;
```

```
void addMatrix(int A[2][2], int B[2][2], int C[2][2]) {  
    for (int i = 0; i < 2; i++) {  
        for (int j = 0; j < 2; j++) {  
            C[i][j] = A[i][j] + B[i][j];  
        }  
    }  
}
```

```
void subtractMatrix(int A[2][2], int B[2][2], int C[2][2]) {  
    for (int i = 0; i < 2; i++) {  
        for (int j = 0; j < 2; j++) {  
            C[i][j] = A[i][j] - B[i][j];  
        }  
    }  
}
```

```
void strassenMultiply(int A[2][2], int B[2][2], int C[2][2]) {  
    int P1, P2, P3, P4, P5, P6, P7;
```

```
    P1 = A[0][0] * (B[0][1] - B[1][1]);  
    P2 = (A[0][0] + A[0][1]) * B[1][1];  
    P3 = (A[1][0] + A[1][1]) * B[0][0];  
    P4 = A[1][1] * (B[1][0] - B[0][0]);  
    P5 = (A[0][0] + A[1][1]) * (B[0][0] + B[1][1]);  
    P6 = (A[0][1] - A[1][1]) * (B[1][0] + B[1][1]);  
    P7 = (A[0][0] - A[1][0]) * (B[0][0] + B[0][1]);
```

```
    C[0][0] = P5 + P4 - P2 + P6;  
    C[0][1] = P1 + P2;  
    C[1][0] = P3 + P4;  
    C[1][1] = P1 + P5 - P3 - P7; }
```

```
int main() {  
    int A[2][2] = {{1, 2}, {3, 4}};  
    int B[2][2] = {{5, 6}, {7, 8}};  
    int C[2][2];
```

```
    strassenMultiply(A, B, C);
```

```
    cout << "Strassen's Algorithm Multiplication Result:" << endl;  
    for (int i = 0; i < 2; i++) {  
        for (int j = 0; j < 2; j++) {  
            cout << C[i][j] << " ";  
        }  
        cout << endl;  
    }
```

```
    return 0; }
```